JGi JAIN FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science & Engineering

Global Campus, Jakkasandra Post, KanakapuraTaluk, Ramanagara District, Pin Code: 562 112

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PCL Attendance

Team No. : COEPCL0014

Project Title : Deep learning frame work for Lung Cancer detection and characterization

Sl. No.	Date of the Meeting	Points Discussed	Members Absent (Roll No)	Signature of the Mentor
1.	March 19th	went through the methodology discuss about drafting the report	Nil	
2.	March-24th	Drafted the report for ~~correct~~ ~~feed~~ feedback for corrections	Nil	
3.	March-30th	make some changes in the report	Nil	
4.	April-8th	Got accurate methodology and reports	Nil	
5.	April-16th	prafted a final report ready for plagrizem check.	Nil	

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PROJECT CENTRIC LEARNING (PCL) PHASE 2

PROJECT TEAM NO.	14

PROJECT TITLE	A Deep Learning Frame Work For Early Lung Cancer Detection And Characterization

TEAM MEMBERS

S. No.	USN No.	NAME OF THE STUDENT	CLASS	MOBILE NO.
1	23BTRCL173	C.NIVAS KRISHNA	AIML-A	8328288043
2	23BTRCL010	D.NAVADEEP	AIML-A	9398730205
3	23BTRCL181	D.HARITHA	AIML-A	9966128168
4	23BTRCL190	G.VENKATA SUJAY	AIML-A	8519990754
5	23BTRCL178	D.GOKUL KRISHNA	AIML-A	9381575192
6	23BTRCL169	B.V.SAI AKSHAYA	AIML-A	9014802628

DETAILS OF THE PROJECT GUIDE

NAME	ARCHANA SASI
DESIGNATION	ASSISTANT PROFESSOR
DEPARTMENT	AIML
eMail ID	archana.sasi2k8@gmail.com
Phone No.	9400284847

DOMAIN OF THE PROJECT

A Deep Learning Framework for Early Lung Cancer Detection and Characterization

Scope of the Project

1. Artificial Intelligence (AI)
The project uses AI to simulate human decision-making.
Helps in automating medical diagnosis tasks.

2. Machine Learning (ML)
The system learns patterns from CT scan data.
Improves performance based on training data without explicit programming..

3. Deep Learning (DL)
Core domain of the project.
Uses Convolutional Neural Networks (CNNs) for image analysis and classification

4. Computer Vision
Deals with understanding and analyzing images.
Extracts features like shape, texture, and patterns from lung scans.

PROJECT GUIDE **PROJECT COORDINATOR**

Existing System:-

Manual Diagnosis by Radiologists
Doctors analyze CT scan or X-ray images manually.
They look for abnormalities like lung nodules.
Requires high expertise and experience.

Traditional Image Processing Techniques
Uses basic image processing methods:
Edge detection
Thresholding
Segmentations.

Classical Machine Learning Models
Algorithms used:
Support Vector Machine (SVM)
K-Nearest Neighbors (KNN)
Random Forest

The existing system relies on manual analysis of CT scans and traditional machine learning methods, which are time-consuming, require manual feature extraction, and are prone to human error, resulting in lower accuracy.

Problem Identification

Late Diagnosis of Lung Cancer

Symptoms appear in advanced stages.
Early-stage detection is difficult but critical for survival..

Manual Analysis of CT Scans

Radiologists must examine large volumes of images.
The process is time-consuming and labor-intensive.

Human Error and Variability

Diagnosis depends on expert interpretation.
Small nodules or subtle patterns may be missed..

Increasing Workload in Healthcare

Growing number of medical images.
Limited number of specialists.

Proposed System:-

Deep Learning-Based Image Classification
The proposed system uses Convolutional Neural Networks (CNN) to automatically classify waste images into different categories .

Transfer Learning with Pretrained Models
The system implements transfer learning using pretrained models like ResNet (ResNet50) and DenseNet121,VIT-Transformer, which are trained on large datasets to improve performance.

Fine-Tuning for Domain-Specific Learning
The top layers of the pretrained models are modified and fine-tuned to adapt specifically to the lung cancer classification dataset, improving accuracy.

Software-Based Implementation
The entire system is implemented as a software solution where users can upload an image, and the model predicts the lung cancer instantly.

Performance Evaluation Using Accuracy Metrics
The system evaluates model performance using metrics such as accuracy and loss to ensure reliable and efficient classification.

No. of Modules

Sl. No.	Name of the Module	Remarks
1	Research & Planning	Research & Planning Literature review, defining objectives & scope of text summarization
2	Data Collection	Collecting datasets (from google and kaggle)
3	Model Selection & Training	CNN-based architectures, specifically DenseNet-121,VIT-Transformer and ResNet50, were selected using transfer learning for their strong feature extraction capability and high performance in image classification tasks.
4	Model Optimization	fine-tuning optimizes the pretrained model by updating selected layers to improve accuracy and adapt it to our specific waste dataset.
5	Testing & Deployment	The trained DENSENET-121,VIT-Transformer and ResNet50 models were tested using evaluation metrics such as accuracy and confusion

		matrix, and the optimized
		model was deployed in a real-time lung cancer prediction.

PROJECT GUIDE **PROJECTCOORDINATOR**